

Building Cooperative Robot Behavior from Basic Action Modules

An LLM-driven control framework for two cooperating AR4 arms

Jan M. Straub

Supervisor: Prof. Dr. Artur Andrzejak

July 29, 2026

Institute for Computer Science, Artificial Intelligence for Programming
Ruprecht-Karls-University of Heidelberg

1 The Problem

The reasoning–execution gap

2 Pillar 1: Grounding

An operation vocabulary makes a small, local VLM reliable

3 Pillar 2: Cooperation

Handoffs and parallelism composed by the model

4 Boundary

The failing task

5 Results

What the ablations really show, limitations

6 Conclusion

Contributions, future work

Condensed Results

Thesis in one sentence

A **locally hosted vision-language model (VLM)**, grounded in a structured pool of **20 basic operations**, turns natural language into **reliable** robot behavior, with **no fine-tuning**.

Solved

Success on all single-robot tasks and dual-arm handoffs with the default backend.

Composed

Cooperation is composed at runtime from general primitives.

Gap

Concurrent bimanual lift.

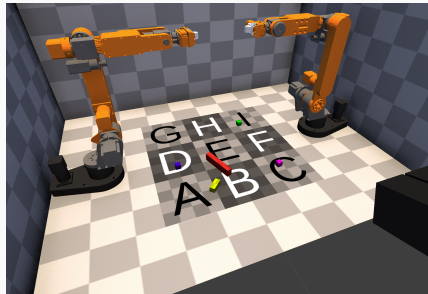
The Problem

The Reasoning–Execution Gap

- Classical robots: precise, but caged. Rigid paths, no tolerance for changes.
- VLMs bring world knowledge, but cannot reliably turn reasoning into *physical* action.
- Models often ignore past physical state when planning the next move.

Why cooperation is the hard case

Two arms must **avoid collision**, **share a workspace**, and **synchronize timing**.



Two AR4 [1] arms, a shared workspace, and a stereo camera in Unity [2].

Background

- **End-to-end VLA** (RT-2) [3]: great generalization, but data-hungry and opaque.
- **Code-as-Policies** [4]: flexible and transparent, but tied to hand-build API set.
- **Framework-scale** (AutoRT) [5]: practical scaffolding, but no joint reasoning over a *shared* object.
- **RoCo** [6]: per-robot agents negotiate, but free-form way-points.

Research gap

Two independent arms compose object coordination from a fixed *grounded* vocabulary of operations, on a local compact model.

Two Design Commitments

Run the model *locally*

- No cloud cost, no external dependency.
- No sensitive scene data leaving the machine.
- A compact VLM is enough.

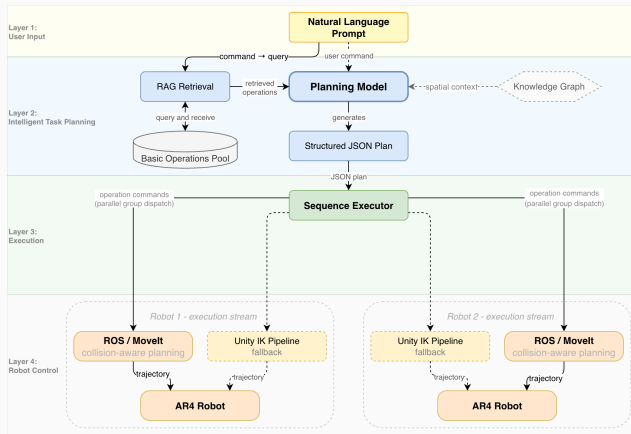
Ground the model in *modular operations*

- Reasoning restricted to a vocabulary of *executable* actions.
- Separates semantic understanding from physical execution.

Default backend model: **Magistral Small 2509 (24B, 4-bit) [7]**

Pillar 1: Grounding

System Architecture



Built on Unity, ROS 2 [8], and MoveIt [9].

The Operation Pool

Operations organized by **complexity**, four tiers:

Atomic (8) indivisible, no perception: gripper, signal, wait.

Basic (6) single dispatch, introduces perception.

Intermediate (4) composite single-robot: placement, stereo detection, scene analysis.

Complex (2) full pipelines: grasp planning, handoff reception.

Why it grounds: model can only express plans the robot can run.

Operation Registry Map (20 operations)

Operation complexity						
	perception	navigation	manipulation	state_check	coordination	sync
			1. grasp_object		1. receive_handoff	
	1. analyze_scene 2. detect_object_stereo		1. place_between_objects 2. place_object			
basic	1. detect_all_fields 2. detect_field 3. generate_point_cloud	1. adjust_end_effector_orientation 2. move_to_coordinate 3. return_to_start_position				
atomic			1. control_gripper 2. release_object	1. check_robot_status	1. yield_workspace	1. reset_simulation 2. signal 3. wait 4. wait_for_signal

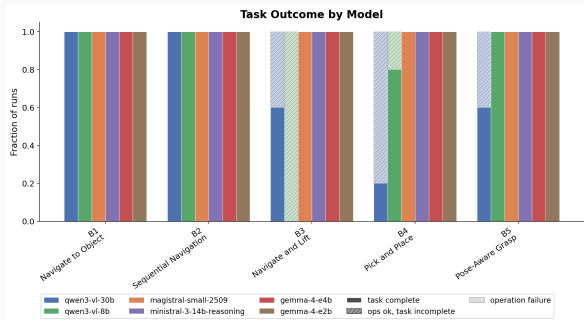
Gated retry on failure

- On an eligible failure, re-query the model with a structured *hint* (Reflexion-style verbal feedback [10]).
- Gated to Navigation / Manipulation / Perception: no point re-querying a sync barrier.

Optional knowledge-graph layer for spatial queries; ablation cost covered later.

Grounding $\Rightarrow$ Reliability

- **B1–B5: success over all 25 runs,** zero hallucinated ops, zero retries.
- *Identical* operation sequence on all 5 runs of each benchmark.



Magistral, Ministral, and both Gemma 4 models clear the full single-robot gradient; the Qwen3-VL models fail from B3 onward by truncating plans.

Pillar 2: Cooperation

The Model Composes

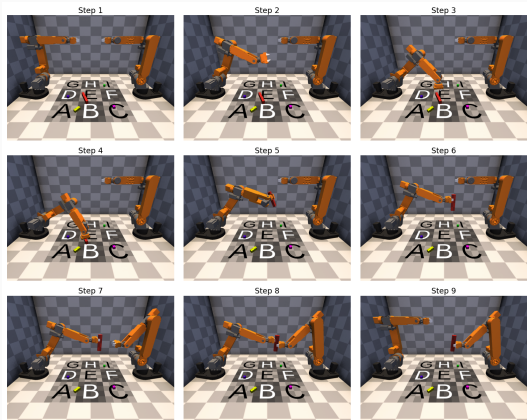
Two general primitives

- `signal / wait_for_signal`: one robot pauses at a checkpoint until the other signals readiness.
- **Parallel groups**: operations dispatched concurrently when no ordering is needed.

The executor has **no** coordination policy.

The Handoff

- Default model: 5/5 runs; all operations first try.
- Ordering and the signal/wait barrier are set by a prompt rule.



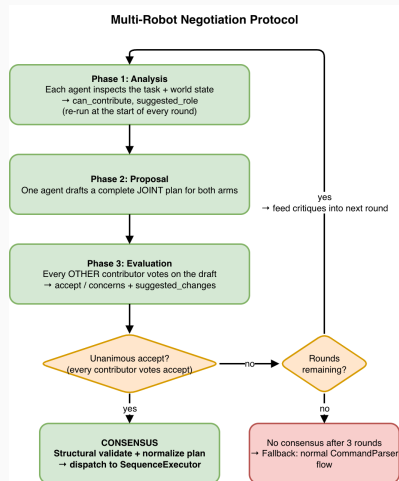
Handoff sequence.

Negotiation

A single model planning both arms under-specifies roles;
implicit conflicts surface only at execution.

Design choices:

- Written accept/reject with justifications, so disagreement is resolved in language, before any motion.
- Proposal rotates round-robin; **unanimity**. A plan omitting any robot is rejected outright.
- The silent fallback keeps negotiation an *option*, not a dependency.



Negotiation Results

	Task SR	Ops/run
Negotiation on	0.713 (57/80)	35.8
Negotiation off	0.613 (49/80)	29.9

- **+10 pp** task success for the **default model**.
- Roughly neutral for three backends; *negative* for two (−19, −21 pp).
- Pays off when the single-shot plan is already close to correct; doesn't rescue weaker planners from role conflicts.

Boundary

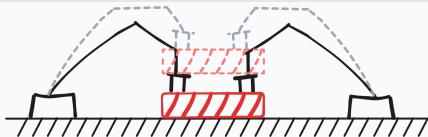
Concurrent Bimanual Lift

All six models fail. Three failure modes:

1. Signal primitives never dispatched once control crosses robots outside a parallel group.
2. Final lift: two *independently* chosen targets land too close together, violating the separation check.
3. Deeper sim limit: a Unity Transform has one parent, so a single object can't attach to two grippers at once.

Gap

Partner-aware target selection. Each arm picks a valid target; nothing checks one *against the other*.

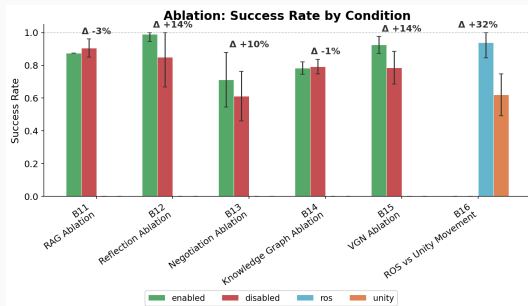


Sketch of the envisioned concurrent bimanual lift.

Results

Ablation Synthesis

- **ROS/Movelt +32 pp**: Unity-only grasp timeouts; plus collision-aware planning.
- **Reflection +14 pp**: clears boundary-reach failures; doubles time when it fires.
- **Negotiation +10 pp**: default model only; neutral/negative on the other five.
- **VGN [11] +14 pp** (+35 pp on grasp-held): center-of-object grasp, not predicted orientation.
- **RAG / KG: net cost**: prompt over-anchoring, helps no backend.



Per-component deltas (default model; $n = 20$ for B11–B13, B16; $n = 5$ for B14–B15).

Limitations

Sensing

Cube features too small to match, so the *dense* stereo cloud is sparse; per-object localization stays robust.

Latency

VLM parse ≈ 20 s median prevents *reactive* control; re-plan between motions, never during.

Sim-to-real

Robot joints and grasp-force thresholds tuned for sim physics, so they need revalidation on hardware.

Conclusion

Five Contributions

1. **Grounded Control Framework:** A hierarchical 20-operation registry, retrieved on demand, turning language into verified behavior with no fine-tuning.
2. **Runtime-Composed Coordination:** The model *sequences* handoff steps and parallelizes independent tasks with `signal/wait` and `parallel` groups.
3. **Capability-Gated Negotiation:** Helps only already capable planners, while degrading weaker models.
4. **Operation-Gated Reflection:** Uses semantic repair to fix physical execution errors.
5. **Two-Layer Safety Adaptation:** Adapts AutoRT [5] to fixed-base manipulators by pairing an LLM semantic constitution with a hard kinematic gate the model cannot override.

Deployment

- `-env real` swaps camera adapter at the abstraction boundary.
- Control switches to ROS/MoveIt collision-aware planning, already configured for dual-robot operation.

Remaining work

- **Depth:** LIDAR or per-robot stereo unlocks VGN-grade grasping.
- **Bimanual:** partner-aware target selection closes the boundary gap.
- **Faster inference:** distilled task-specific models or speculative decoding to shrink the 20 s window.

Demo & Questions

What works

- Single-robot: 25/25 runs.
- Long chain: 85 ops/run, 5 cycles, no cascading errors.
- Handoff: works; sync barrier correctly realized.
- Parallelism: works for 4/6 backends.
- Safety gate: no false-accept, reproducible.

What each component contributes

- VGN grasp: +14 pp op-SR (+35 grasp-held).
- ROS/Movelt: +32 pp, Unity-only grasp timeouts.
- Reflection: +14 pp task.
- Negotiation: +10 pp, default model only.
- RAG: net cost, helps no backend; KG: small cost.

The one open gap

- Concurrent bimanual B7: all six models fail.
- Causes: mis-realized sync barrier; targets not partner-aware; single-parent attach.

Thank you.

`github.com/JanMStraub/Cooperative-Robot-Behavior-from-Basic-Action-Modules`

References

- [1] C. Annin, "Annin robotics," Accessed: 2026-06-09, 2026. [Online]. Available: <https://anninrobotics.com>
- [2] Unity Technologies, "Unity," 2026, game development platform. [Online]. Available: <https://unity.com/>
- [3] A. Brohan, N. Brown, J. Carbajal et al., "Rt-2: Vision-language-action models transfer web knowledge to robotic control," 2023. [Online]. Available: <https://arxiv.org/abs/2307.15818>
- [4] J. Liang, W. Huang, F. Xia et al., "Code as policies: Language model programs for embodied control," 2023. [Online]. Available: <https://arxiv.org/abs/2209.07753>
- [5] M. Ahn, D. Dwibedi, C. Finn et al., "Autort: Embodied foundation models for large scale orchestration of robotic agents," 2024. [Online]. Available: <https://arxiv.org/abs/2401.12963>
- [6] Z. Mandi, S. Jain, and S. Song, "Roco: Dialectic multi-robot collaboration with large language models," 2023. [Online]. Available: <https://arxiv.org/abs/2307.04738>
- [7] Mistral AI, "Magistral-small-2509," <https://huggingface.co/mistralai/Magistral-Small-2509>, 2025, large Language Model.
- [8] S. Macenski, T. Foote, B. Gerkey et al., "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, May 2022. [Online]. Available: <http://dx.doi.org/10.1126/scirobotics.abm6074>
- [9] D. Coleman, I. Sukan, S. Chitta et al., "Reducing the barrier to entry of complex robotic software: a moveit! case study," 2014. [Online]. Available: <https://arxiv.org/abs/1404.3785>
- [10] N. Shinn, F. Cassano, E. Berman et al., "Reflexion: Language agents with verbal reinforcement learning," 2023. [Online]. Available: <https://arxiv.org/abs/2303.11366>
- [11] M. Breyer, J. J. Chung, L. Ott et al., "Volumetric grasping network: Real-time 6 dof grasp detection in clutter," 2021. [Online]. Available: <https://arxiv.org/abs/2101.01132>

Additional Slides

Backup: Research Questions

- RQ1 (Grounding)** Can a locally-hosted VLM, grounded in a retrieval-backed pool of verified operations, reliably decompose natural-language tasks into executable plans without fine-tuning?
- RQ2 (Cooperation)** Can that **same** grounding let the model compose dual-arm coordination at runtime?
- RQ3 (Boundary)** Where does it break?

Backup: Bimanual Lift per-run Outcomes

Run	Ops ok	Dur (s)	Failing op
1	4/8	111	wait_for_signal
2	4/10	43	wait_for_signal
3	6/7	211	move_to_coordinate (collision)
4	6/7	206	move_to_coordinate (collision)
5	4/10	46	wait_for_signal

Two clusters: mis-realized barrier, never executed (1,2,5), and lift-target collision at ~ 0.10 m (3,4).

Backup: Prompt-Driven Parallelism

General rule in the prompt

```
Same parallel_group = concurrent.  
Later group waits for all prior.  
VAR DEPENDENCY LAW: if B reads a  
$var captured by A, B must be in a  
strictly higher parallel_group.
```

B10 plan the model emitted

```
g1: detect(R1,blue) || detect(R2,green)  
g2: grasp(R1) || grasp(R2)  
g3: field(R1,A) || field(R2,B)  
g4: place(R1) || place(R2)
```

- 4/6 models: 40/40 operations, zero retries.
- Each grasp waits only on its own detection (\$var dependency); the two arms otherwise run fully in parallel.

Backup: AutoRT Adaptation & Safety Gate

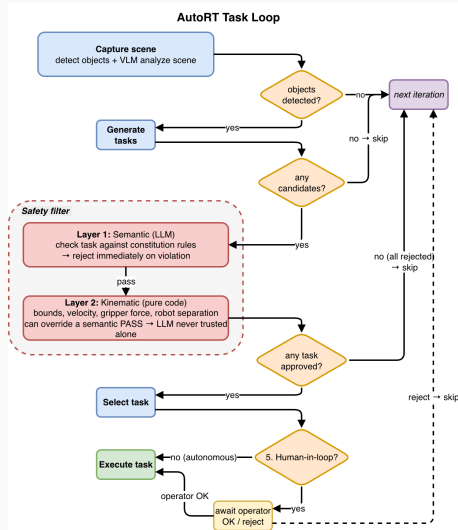
Re-targeted AutoRT for *fixed-base* manipulation.

Two-layer safety (Robot Constitution):

- Semantic VLM check
- Hard kinematic check the semantic layer can't override.

Result: never admits an unsafe task.

Slightly over-rejects aggressive-but-safe phrasing.



Backup: What VGN changes

Held & lifted	VGN on	VGN off
Magistral	14/20 (0.700)	7/20 (0.350)
Pooled, 6 models	0.750	0.417

- Every backend improves: +20 pp (Ministral) to +65 pp (Gemma-4B).
- Success = held and lifted clear of the table.

Mechanism: position, not orientation

- **Same orientation both conditions:** top-down, yaw from the object's long axis. VGN's own poses fail motion planning and are discarded.
- **The difference is position:** VGN grasps the detected object *center*; the baseline's contact point sits off-center on angled cubes and slips.

Scope

Position is the stereo-written WorldState estimate, so B15 isolates *grasp positioning*. Recovering VGN's learned pose needs a real depth sensor.

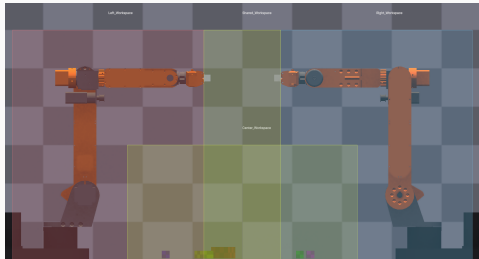
Backup: Who Writes the World State

Field	Writer	Note
Robot state	Unity broadcast	pose, joints, gripper; the simulation <i>is</i> the robot
Object position	detect_object_stereo	stereo estimate; Unity never overwrites it
Dimensions, orientation	Unity (collider)	no perception producer yet; the sim-to-real gap

- Confidence decays once an object leaves view, stale entries expire, and a background refresh re-detects them.
- So the grasp reaches a *perceived* position; what simulation still supplies is dimensions and orientation.

Backup: Collision Safety Layers

1. **Static pre-check** (Python, before dispatch): checks workspace bounds, object conflicts, and deadlocks on the planned targets.
2. **Workspace regions**: left / right / shared zones; a contested zone is handed over explicitly via `yield_workspace`.
3. **Runtime guard** (Unity, every physics step): ProximityGuard freezes motion at 0.25 m end-effector distance, resumes at 0.35 m; link-to-link limit 0.15 m.



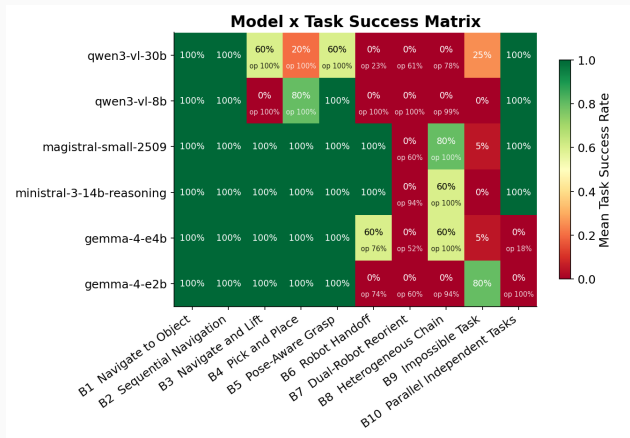
The LLM is never trusted for safety: layers 1 and 3 are code, not reasoning.

Backup: Single-Robot Operation Latencies

Operation	Bench	n	Mean (s)
detect_object_stereo	B1–B5	30	0.80
move_to_coordinate	B1–B3	20	4.37
grasp_object (B3)	B3	5	19.84
grasp_object (B4)	B4	5	10.93
place_object	B4	5	34.43

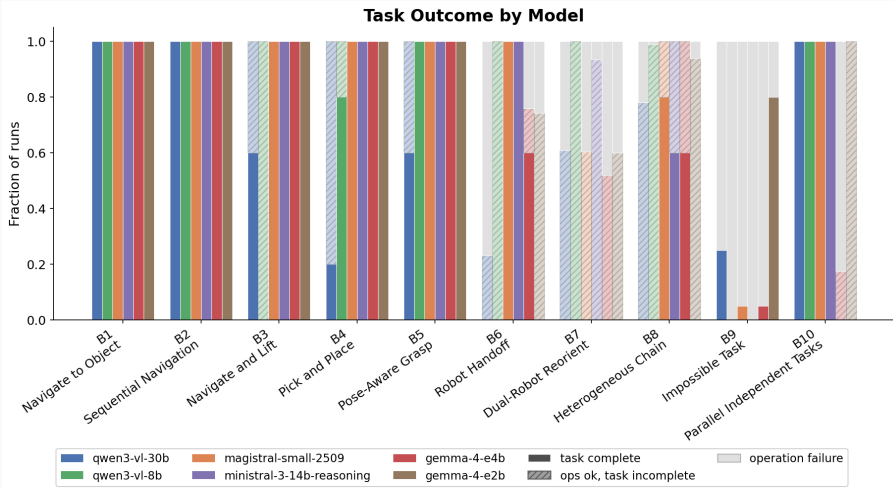
Perception is cheap and stable; contact-rich manipulation dominates and varies with object pose.

Backup: Model × Task Matrix



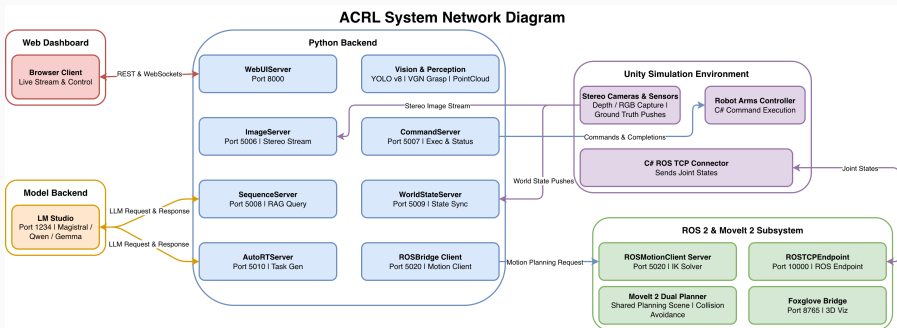
Task-level success rates across all six VLM backends for benchmarks B1–B10, highlighting single-robot completion and multi-robot capability separation.

Backup: Complete Success Rate Graph



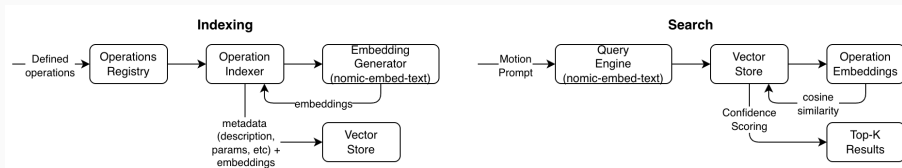
Comparative success rate breakdown per benchmark tier across all six evaluated models (five runs per condition).

Backup: Network Map



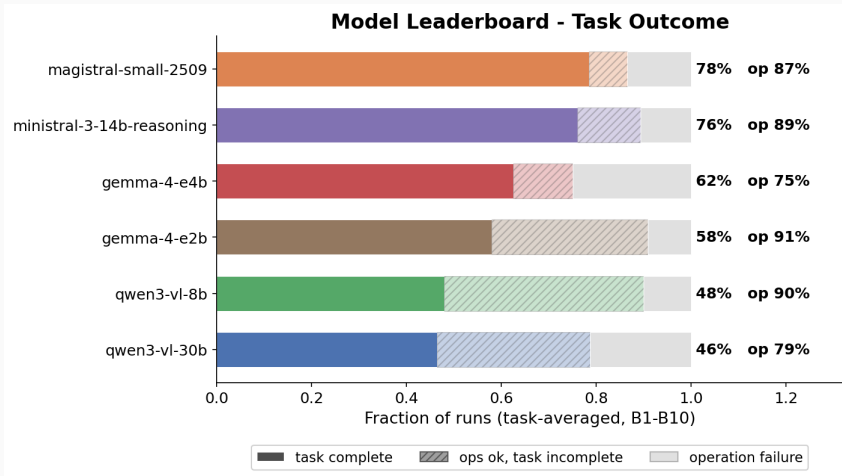
Distributed system architecture showing TCP socket bridges connecting the Python sequence executor, Unity simulation, WorldState server, and ROS 2/MoveIt stack.

Backup: RAG



Retrieval-Augmented Generation workflow: converts natural language instructions into query embeddings to retrieve top-k operations and workflow exemplars for prompt assembly.

Backup: Model Leaderboard



Comprehensive model ranking based on task completion rate, operation execution reliability, and plan formatting conformance.